



EAST WEST UNIVERSITY

“Project Report”

Course Title: Statistics for Data Science

Based On: “Student Depression Dataset”

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1. Introduction:

Depression among students is now a worldwide phenomenon, as this report shows. It affects quality of life, academic performance, and general well-being. As the pressures of education grow, students are under pressure from many directions. They must meet academic demands, keep up with work commitments, and avoid the strain of financial stress if they are to be free to learn. In addition, relationships in personal life inevitably suffer. This report is trying to analyze factors contributing to student depression and find a model that can predict which students may suffer from such depression. The data used for analysis contains various features concerning students' personal information, academic performance, lifestyle behaviors, and history of mental health. So, this report aims to discover more about these adverse relationships, how the various attributes of student life influence depression risk and finally aims at using attributes to build a machine learning model predicting students' probability for this disorder. This model can help detect who is at risk of depression in good time, making intervention possible promptly when necessary.

2. Description of the Dataset:

The dataset contains 27,901 rows and 18 columns detailing students' psychological well-being and academic experiences. Each line represents a student, and the columns store features composed of their background data, educational history, lifestyle information, and psychological health records. The dataset columns are as follows:

ID: The number of the row in the data set reflects a unique identity for each student. This is an integer value that separates individual records.

Sex: A Category data item representing the sex of students (Male/Female).

Age: Age of students. This continuum is a numerical variable.

City: The city where a student resides. This is a category data item. Since different cities exist, geographical trends in depression may also exist.

Profession: The student's profession, being a 'Student' or another job. In this dataset, all the columns will likely be labeled as 'Student.'

Academic Pressure: A measured continuous numerical expression of students' academic pressure. This feature provides a window into what kind of pressure school levels can put on an individual.

Work Pressure: Students face several work-related pressures. This may cause him to feel a general state of stress and thus possibly lead to depression.

GPA: A student's actual grade point average reflects their academic record.

Study Satisfaction: Measured on a numerical scale the degree to which students are satisfied with their studies or course.

Job Satisfaction: The degree of student satisfaction in their job, if applicable.

Sleep Duration: This is the student's sleep period. It is categorical data, such as 5-6 hours or 7-8 hours.

Dietary Habits: The category of food students typically eat (Health, Moderate).

Degree: The name of the degree the student pursues (e.g., B.Sc, M.Tech).

Suicidal Thoughts: A binary variable showing whether or not students have experienced thoughts of suicide (Yes/No). This is important for understanding the extent of mental health problems.

Work/Study Hours: Student's total hours spent on work and study.

Financial Stress: A numerical variable indicating the level of financial strain experienced by students.

Family History of Mental Illness: A binary data that shows whether students come from a family with a history of mental illness. (Yes/No)

Depression: The target variable shows if a student is depressed (1 for Yes, 0 for No). With its large data amount (27,901 rows), this dataset offers a solid grounding for robust statistical analysis and predictive modeling. That enables one to dig deep and discover, for example, what various factors cause student depression.

3. Data Preprocessing:

Data preprocessing is a critical approach for converting the raw dataset into a clear, organized form that is appropriate for analysis and modeling. Some automated procedures were executed beforehand to ascertain data's quality and preparatory development for machine learning. These tasks include dealing with missing values, encoding categorical variables, scaling features, and preparing the data for model training.

3.1 Handling Missing Values:

Before encoding the categorical variables, we should check for and deal with missing values. For instance, missing data can bias the analysis results and predictions provided by our and similar methods.

City: Categorical column indicating the city in which the student lives. In this situation, we can replace any missing value with the mode (the most common city).

Financial stress: Some values might also be missing from this numeric column. We will fill in the NaNs in this column with the average value for the column.

The rows will be dropped for the columns with fewer missing data, or they will be imputed by mean/mode/median, depending on the type of column.

3.2. Encoding with Categorical Mapping:

For non-numeric variables, categories are mapped to numeric. This guarantees that the model can understand the categorical variables.

Gender: Sex is a binary variable that takes "Male" and "Female". We can translate these into numbers as:

- Male = 0
- Female = 1

City: The city field has more than one city name. Mapping with which we map all unique cities to a unique integer. For example:

- Visakhapatnam: 1,
- Bangalore: 2,
- Srinagar: 3,
- Varanasi: 4,
- (and so on for other cities)

Profession: The Profession field, which is probably always "Student" for each row, can be interpreted as:

- Student: 1,
- Civil Engineer: 2,
- Architect: 3,
- (and so on for other Professions)

Sleep Duration: We have ordinal values in the Sleep Duration column, such as '5-6 hours', '7-8 hours', and so on. This will be transformed into the following:

- Less than 5 hours: 0,
- 5-6 hours: 1,
- 7-8 hours: 2,
- More than 8 hours: 3,
- Others: 4

Family history of a member with mental disease: It can be modeled as the binary variable:

- No = 0
- Yes = 1

Have you ever thought of killing yourself? This binary variable was represented as:

- No = 0
- Yes = 1

Degree: Degree can be any type, such as Class 12, B.Sc, M.Tech, BA, etc. Every degree will be directly mapped to an integer:

- For Class 12 = 1

- For bachelor's all Degree = 2.
- For master's all Degree = 3.
- Others = 4

Dietary Habits: The column includes categories that show "Healthy" and "Moderate." These will be mapped as follows:

- Healthy: 0,
- Moderate: 1,
- Unhealthy: 2,
- Others: 3

3.3. Encoding the Target Variable:

The response, Depression, is binary (0: non-depressed, 1: depressed). Thus, it was already in an appropriate format for classification problems. Still, we had to balance the distribution of class (depressed/non-depressed) before starting the model with actual training of the model. If there is a class imbalance, other strategies, such as oversampling the minority class or adjusting class weights in the model, can also be used.

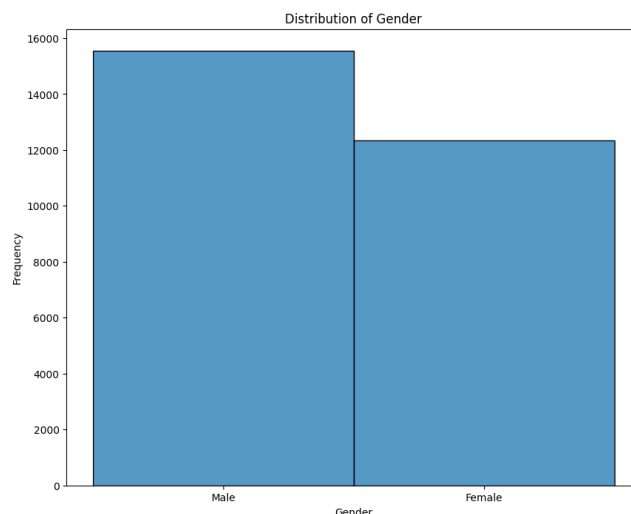
3.4. Data Splitting:

To evaluate the performance of the ML models, we divided the dataset into training and testing subsets. An ordinary split of the data, 80% training, and 20% test were adopted. The training set was used for model training, and the test set was kept aside to check the model's performance on unseen data. In addition, cross-validation was used when selecting the models to make them robust.

4. Exploratory Data Analysis (EDA):

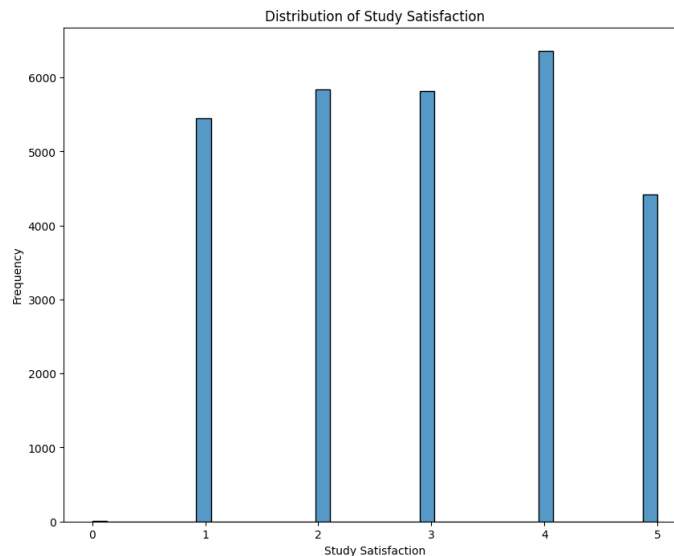
4.1. Histogram Plots

Distribution of Gender:



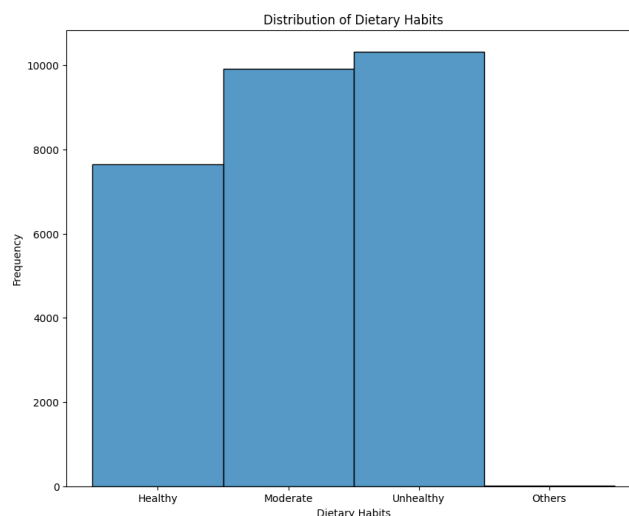
Judging from the number of appearances, males occur in much greater quantity overall than females. The male bar is higher than that of the female bar, so there are more males in the dataset.

Distribution of Satisfaction with the Studies:



The proportion of study satisfaction ranges from 1 to 5, with its distribution almost identical over its range, as most of the responses are concentrated at 1, 2, 3, and 4. The student satisfaction levels of 0 and 5 are slightly lower, which may also suggest that the theme at the university is excellent.

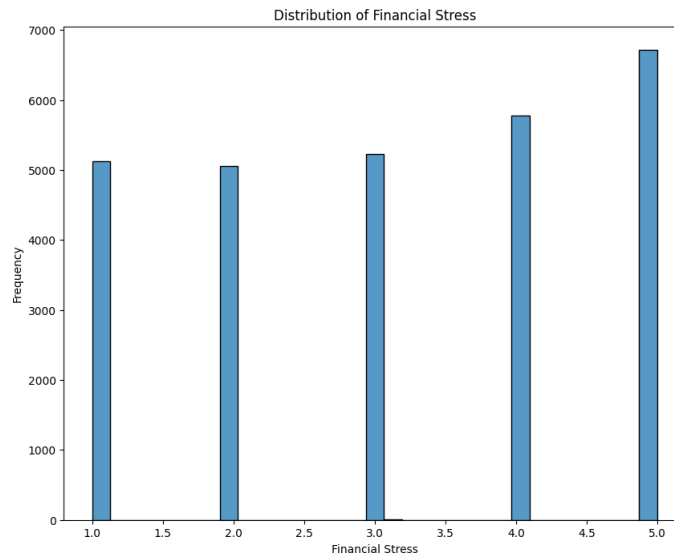
Frequency of Dietary Intake:



We note that most of the population is either the "Moderate Diet" or "Unhealthy Diet," with a slight trend down to the slightly less standard "Healthy Diet" and

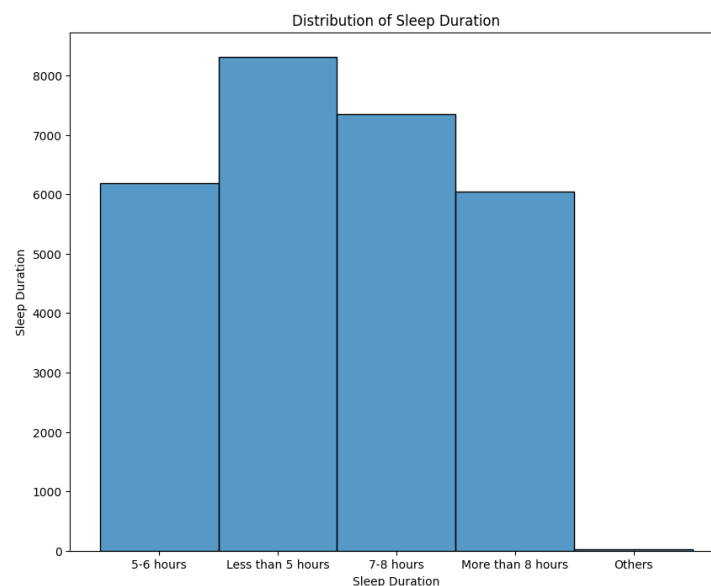
"Others." This means at least a sizeable minority of people are not eating as well as they could.

Dissemination of Financial Stress:



The distributions are even; it seems that we're getting more of a response to higher levels of stress (4 and 5). This suggests that financial stress might be a problem for some of the participants.

Distribution of hours of sleep:

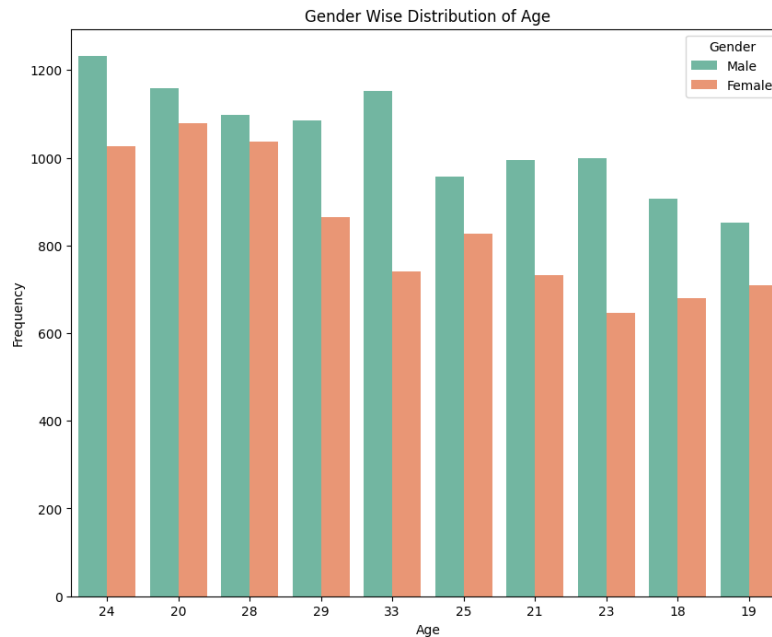


Nearly all individuals report sleeping less than 7 hours, and "Less than 5 hours" is the most common category. This indicates that many people might be depriving

themselves of sleep, and that could be due to bad sleeping habits or just overpacked schedules.

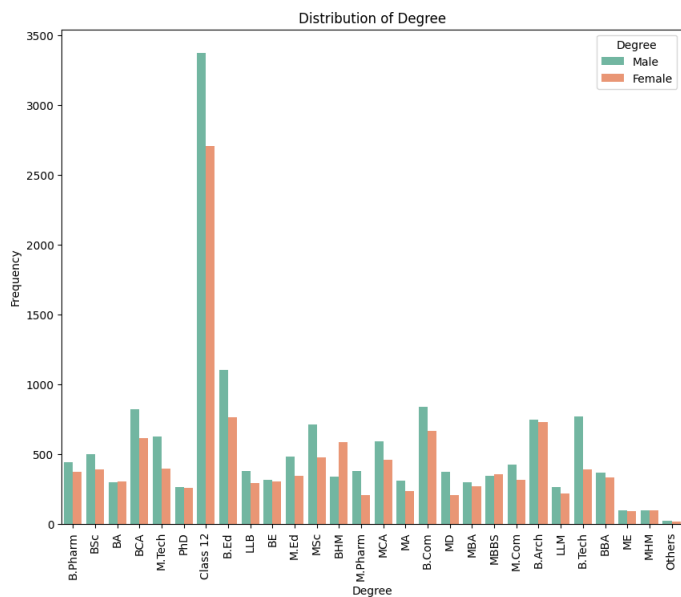
4.2. Count Plots

Age Distribution of Age by Gender:



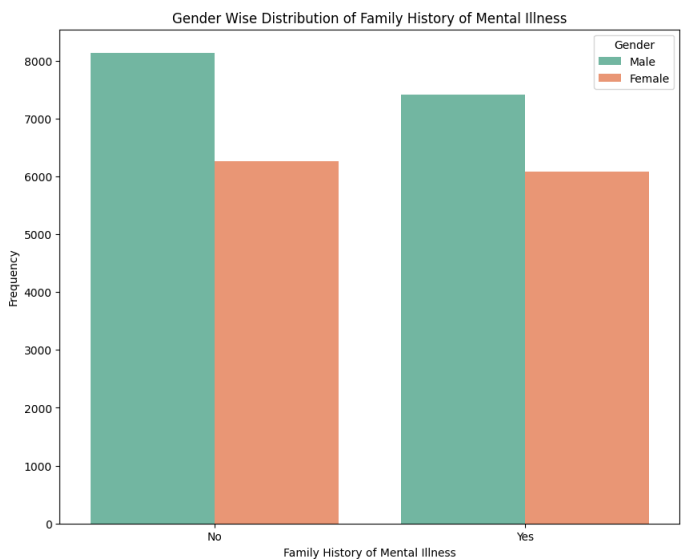
The distribution shows that the number of males is higher than that of females in every age group. In that order, it is most common at age 24, then 20 and 28, with males predominant in all these ages. The trend indicates that most users are male and aged from early to late twenties.

Distribution of Degrees:



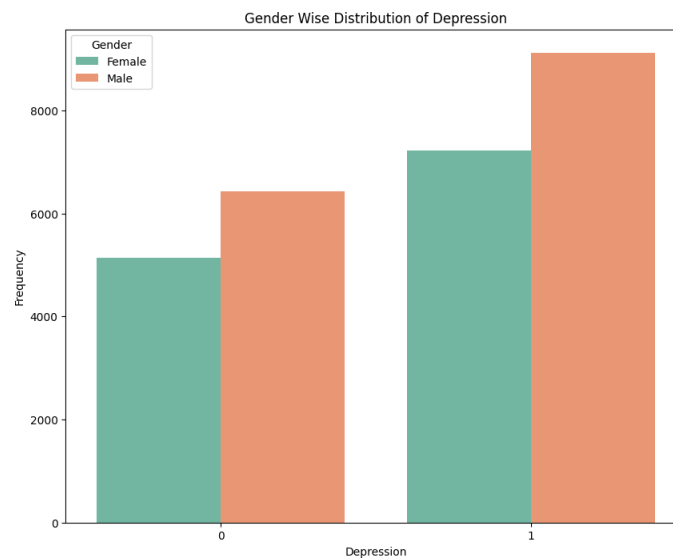
This count plot presents a diversity of educational qualifications. The most common education for both sexes is Class 12, and males are significantly more likely to have it. Men also lead in higher degrees at Ph. D., M.Tech, and B.Tech. Females are present among all types of degrees but are less represented in almost all of them, particularly in the technical and higher education fields. This point indicates a gender discrepancy in higher academic achievement, especially in the technical domains.

Distribution of History of Mental Illness in the Family According to Gender:



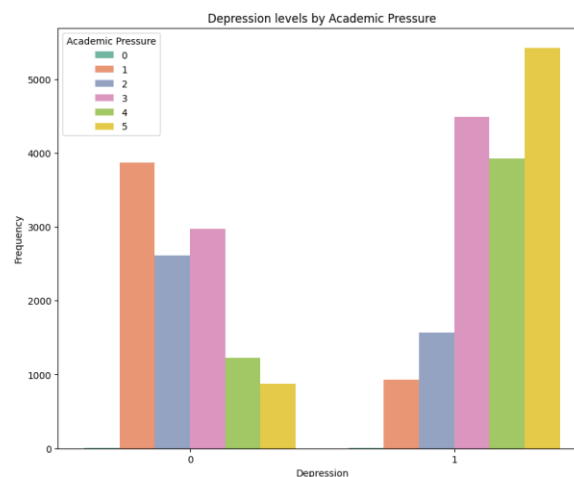
The male sex had more individuals reporting both ‘yes’ and ‘no’ for a family history of mental illness (Resulting in a smaller female-to-male ratio). However, the gender discrepancy, except for age and degree charts, is insignificant. A large percentage of both sexes reported a positive family history, indicating that there is probably a necessity for mental health promotion and support programs in persons who belong to affected families.

Distribution of Depression According to Gender:



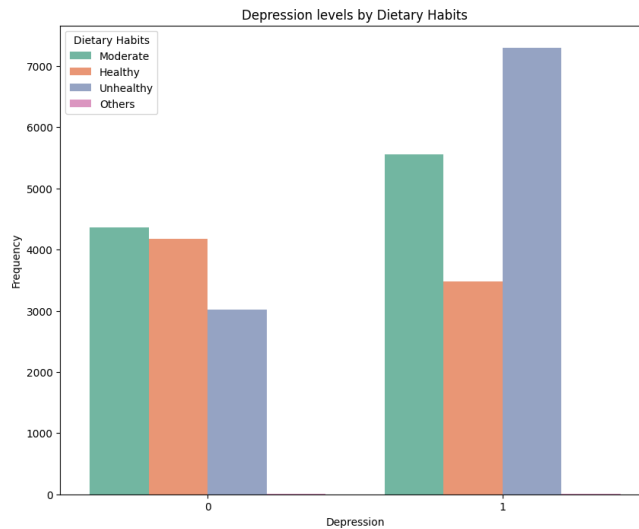
It further shows that there are more men than women in the data. Male depression is more frequent since (in general) there are more depressive males than females.

Depression Levels of Academic Pressure:



It's bad enough; the more pressure in school, the more depressed. Levels 4 and 5 are also significantly more depressed than the lower levels.

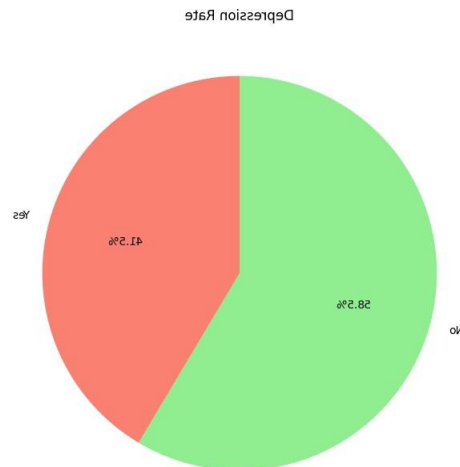
Depression Levels According to Food Preferences:



Poor dietary habits are highly associated with an increased risk of depression. People with better or fair diets have less depression.

4.3. Pie Charts

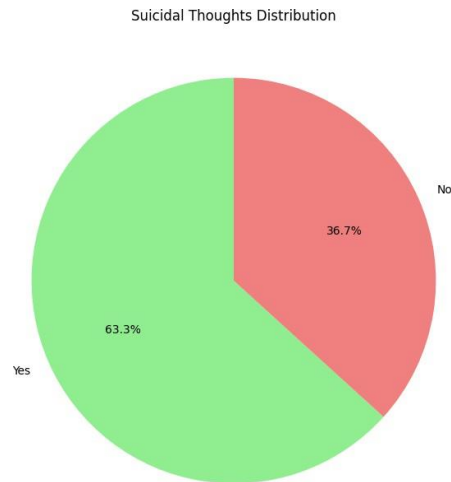
Rate of depression:



41.5% of respondents said they experienced depression. 58.5% indicated that they did not have depression.

Perspective: The rate of depression among the population was above 40%, suggesting a serious mental health problem.

Distribution of Suicidal Thoughts:

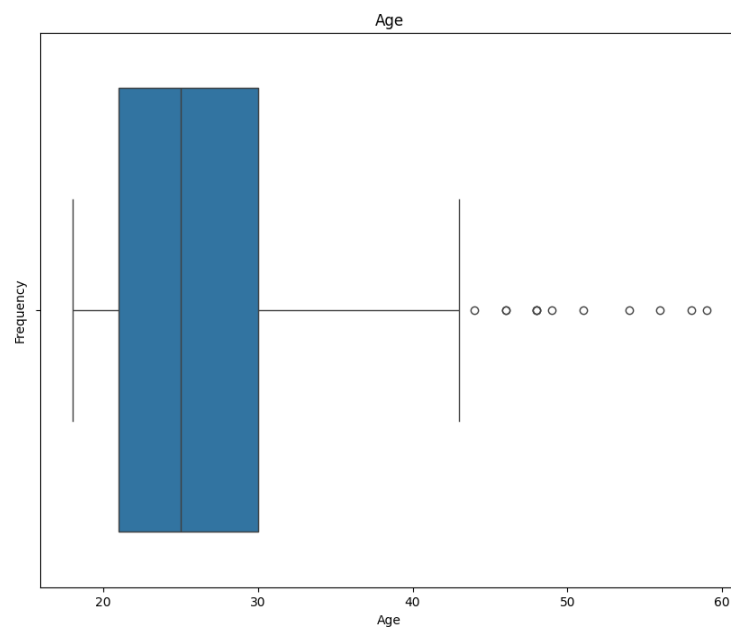


63.3% of them had suicidal thoughts. 36.7% of respondents replied that they do not have suicidal thoughts.

Takeaway: The rate of thinking about suicide — nearly two-thirds of those surveyed indicate that they have — is shockingly high. Let me know if you want to hear more of a crunched comparison/analysis.

4.4. Box Plots

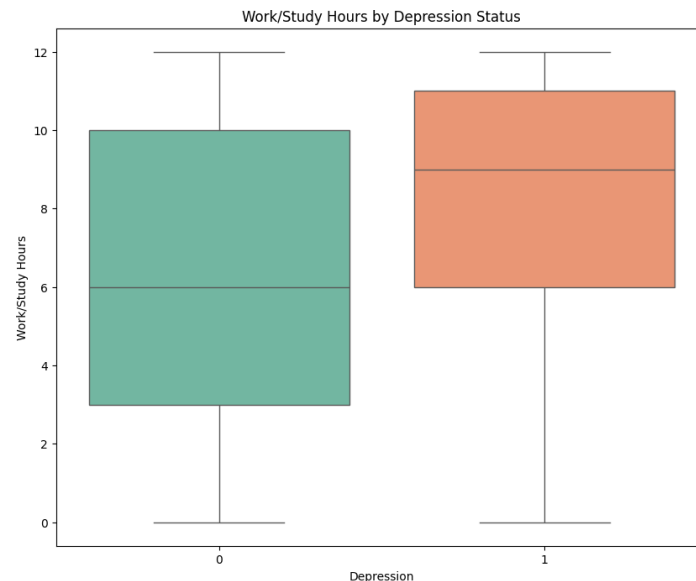
Age Distribution:



The center of age distribution is young adults, which means majority of the users are between 20 and 35 years old. The average age is 25–26. There are some older

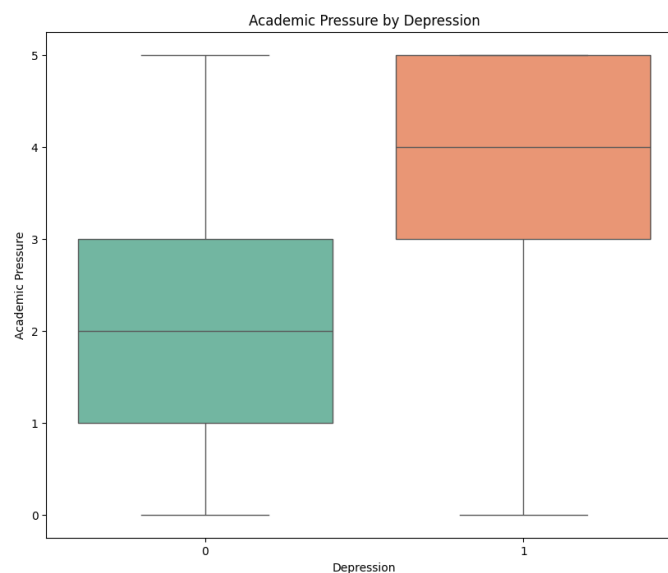
people above 40 which are outliers and thus are less frequent in the dataset. This means the data is mostly for a younger crowd.

Number of Work/Study Hours According to Depression Status:



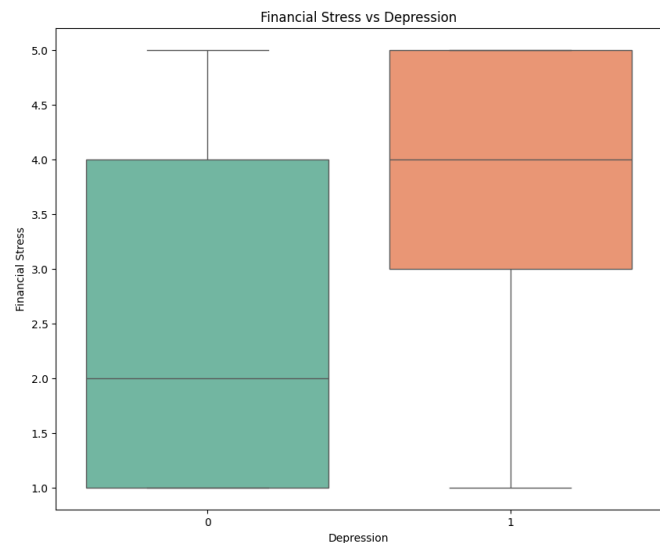
This Graph shows a comparison of the hours worked/studied in the non-depressed group and the depressed group. Respondents not having depression (Depression = 0) present a lower median work/study time and greater dispersion. By contrast, those with depression (Depression = 1), on average, work or study more extended hours, hinting at a possible association between workload and depression.

The Pressure of Academic by Depression:



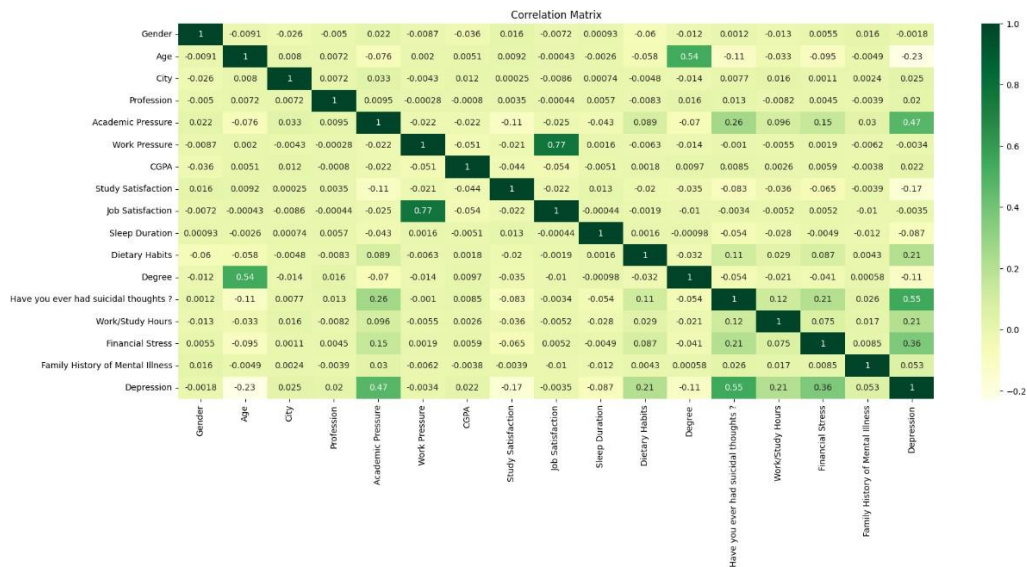
This Graph shows the academic stress reported by subjects by their depression status. Individuals with depression report higher academic pressure, indicated by the higher median and the broader range. This is an indication that academic stress may be a risk factor for depression.

Financial Stress and Depression:



The third Graph looks at financial stress and how it relates to depression. People with depression are likely to report higher financial stress than those without. The skewed distribution of the depression group towards higher stress levels is a pointer to the association between depression and economic constraints.

5. Correlation Matrix



The matrix of correlations indicates the link's strengths among various variables affecting mental health and demographics.

Strong correlations:

Job Satisfaction and Work Pressure (0.77)

Depression and Suicidal Thoughts (0.55)

Negative correlations:

Depression and Sleep Duration: -0.087

Depression and Study Satisfaction: -0.17

In general, **Job satisfaction and Work pressure** are highly correlated, **Depression and Sleep Duration** negatively correlated.

6. Machine Learning Models:

Here is a brief report depending on the classification results using the two algorithms LR(SVM)5 for your data.

Result logistic regression (LR):

Accuracy: 84.6%

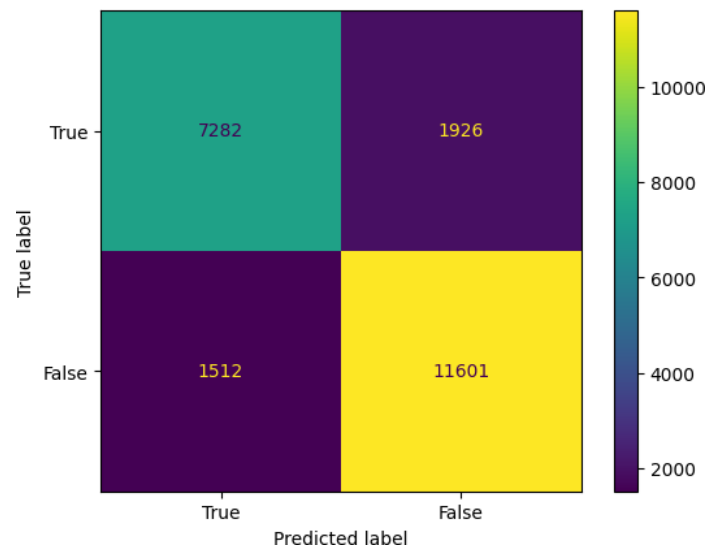
Precision: 85.8%

Recall (Sensitivity): 88.5%

F1-Score: 87.1%

These results indicate that the Logistic Regression model does well with high precision and recall, suggesting that it effectively identifies both positive and negative cases. The F1-Score, a balanced measure, also means that the model is doing well regarding precision and recall.

Confusion Matrix (LR):



Support Vector Machine (SVM) Performance:

Accuracy: 84.6%

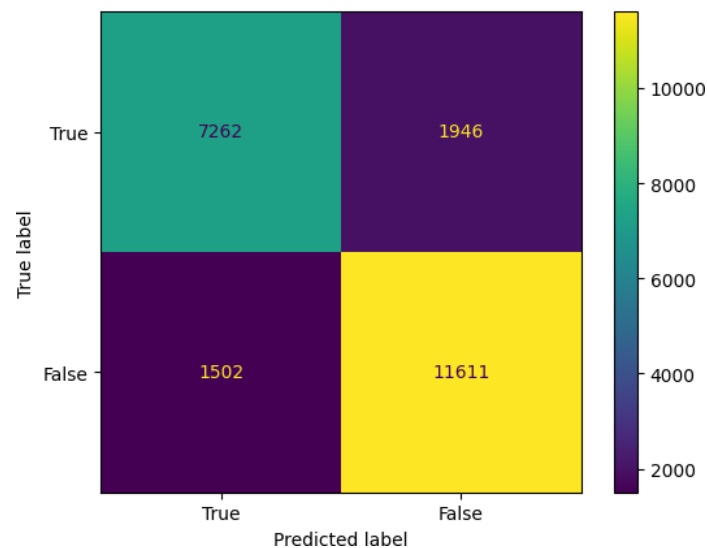
Precision: 85.6%

Recall (Sensitivity): 88.5%

F1-Score: 87.1%

The SVM model performs almost on par with the Logistic Regression model except for precision. This suggests that whether the model is Logic Regression or SVM, the overall performance on this dataset does not alter greatly.

Confusion Matrix (SVM):



7. Conclusion:

Both models can predict the target variable from the dataset, and as such, they could be employed depending on other considerations like training time, model complexity, or interpretability.

From the evaluation metrics, accuracy, precision, recall, and F1-score also do not make a big difference between **Logistic Regression (LR)** vs. **Support Vector Machine (SVM)**.

However, **logistic regression** is less complicated, faster, and more interpretable. If the relationships are linear, this is likely the preferred solution for your dataset.

If you know that there's going to be a nonlinear relationship in the data, we recommend using SVM. However, Logistic Regression, if used on the same, can avoid a huge number of calculations and work very nicely in this case.

Therefore, the **Logistic Regression** is better performance in this dataset.

End